Claim 1: $\hat{\delta} = y_i - \hat{y}_{(i)}$

Claim 2: $t_i := \dfrac{\hat{\delta}}{se(\hat{\delta})} = \dfrac{y_i - \hat{y}_{(i)}}{\hat{\sigma}_{(i)} (1 + x_i^T P_{(i)}^{-1} x_i)^{1/2}}$ (*)

Claim 3: RHS of (*) $= \dfrac{\hat{\varepsilon}_i}{\hat{\sigma}_{(i)} \sqrt{1-h_{ii}}}$ (Sen, Srivastava P157)

Lem 1: (inversion formula for rank-1 update)

$$P_{(i)}^{-1} = (X^T X)^{-1} + \frac{(X^T X)^{-1} x_i x_i^T (X^T X)^{-1}}{1-h_{ii}}$$

proof of Claim 2:

$$\begin{aligned} \mathrm{var}(\hat{\delta}) &= \mathrm{var}(y_i - \hat{y}_{(i)}) \\ &= \mathrm{var}(y_i) + \mathrm{var}(\hat{y}_{(i)}) \\ &= \sigma^2 + \sigma^2 x_i^T (X_{[i]}^T X_{[i]})^{-1} x_i \\ &= \sigma^2 + \sigma^2 x_i^T P_{(i)}^{-1} x_i \end{aligned}$$

$e_i = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \leftarrow$ i-th

$X_{[i]}$: set i-th row = 0

x_i : transpose of i-th row

$P_{(i)} = X_{[i]}^T X_{[i]} \in \mathbb{R}^{p' \times p'}$

$\hat{y}_{(i)}$: leave-one-out prediction

$\hat{\beta}_{(i)} \in \mathbb{R}^{p'}$

$\hat{y}_{(i)} = x_i^T \hat{\beta}_{(i)}$

Proof of Claim 3: it suffices to show that

$$y_i - \hat{y}_{(i)} = \frac{\hat{\varepsilon}_i}{1-h_{ii}} \quad (1)$$

$$1 + x_i^T P_{(i)}^{-1} x_i = \frac{1}{1-h_{ii}} \quad (2)$$

(2) follows from Lem 1. To see (1), note that Lem 1 implies

$$\begin{aligned} x_i^T (X_{[i]}^T X_{[i]})^{-1} X_{[i]}^T &= e_i^T H - h_{ii} e_i^T + \frac{h_{ii} (e_i^T H - h_{ii} e_i^T)}{1-h_{ii}} \\ &= \frac{1}{1-h_{ii}} e_i^T H - \frac{h_{ii}}{1-h_{ii}} e_i^T \end{aligned}$$

$$\Rightarrow \quad e_i^T - x_i^T P_{(i)}^{-1} X_{[i]}^T = \frac{e_i^T (I-H)}{1-h_{ii}}$$

$$\Rightarrow \quad y_i - y_{(i)}^T := y_i - x_i^T P_{(i)}^{-1} X_{[i]}^T y$$

$$= \frac{e_i^T (I-H) y}{1-h_{ii}} = \frac{\hat{\varepsilon}_i}{1-h_{ii}}$$

Proof of Claim 1

$$\mathbb{R}^{p'} \ni \hat{\beta}_{(i)} := \operatorname*{argmin}_{\beta \in \mathbb{R}^{p'}} \sum_{j \neq i} (y_j - x_j^T \beta)^2$$

$$\mathbb{R}^{p'+1} \ni \hat{\beta}_* := \operatorname*{argmin}_{\beta \in \mathbb{R}^{p'},\, \delta \in \mathbb{R}} \sum_{j=1}^{n} \left(y_j - x_{*j}\begin{pmatrix}\beta \\ \delta\end{pmatrix}\right)^2$$

$$= \operatorname*{argmin}_{\beta,\, \delta} \left\{ \sum_{j \neq i} (y_j - x_j^T \beta)^2 + (y_i - x_i^T \beta - \delta)^2 \right\}$$

$$\hat{\beta}_* = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\delta} \end{pmatrix} \begin{matrix} \in \mathbb{R}^{p'} \\ \in \mathbb{R} \end{matrix}$$

$$\begin{cases} \hat{\beta}_{*1} = \hat{\beta}_{(i)} \\ \hat{\delta} = y_i - \underbrace{x_i^T \hat{\beta}_{(i)}}_{\hat{y}_{(i)}} \end{cases}$$

Claim 4: $$r_i \left(\frac{n - p' - 1}{n - p' - r_i^2} \right)^{1/2} = \frac{\hat{\varepsilon}_i}{\hat{\sigma}_{(i)} \sqrt{1 + h_{ii}}}$$

Proof: It suffices to show

$$\frac{1}{\hat{\sigma}} \left(\frac{n - p' - 1}{\cdots} \right)^{1/2} = \frac{1}{\hat{\sigma}_{(i)}}$$

$$\Leftrightarrow \quad \frac{n - p'}{n - p' - r_i^2} = \frac{\hat{\sigma}^2 (n - p')}{\hat{\sigma}_{(i)}^2 (n - p' - 1)}$$

$$\Leftrightarrow \quad \frac{r_i^2}{n-p'} = \frac{-\hat{\sigma}_{(i)}^2 (n-p'-1) + (n-p')\hat{\sigma}^2}{\hat{\sigma}^2 (n-p')}$$

$$\overset{\text{Lemma 2,3}}{=} \frac{\varepsilon V V^T \varepsilon}{\hat{\sigma}^2 (n-p')}$$

$$\Leftrightarrow \quad r_i^2 = \frac{\| \varepsilon^T V \|^2}{\hat{\sigma}^2} = \frac{|e_i (I+H) \varepsilon|^2}{(1-h_{ii})\hat{\sigma}^2}$$ Done!

Lemma 2: Let H' = the projection onto column space of $X_* := [X, e_i]$. Then $H' - H = VV^T$ where $V = \frac{(I-H)e_i}{\|(I-H)e_i\|}$.

Lemma 3: we have

$$\hat{\sigma}^2 = \|(I-H)\varepsilon\|^2$$

and $$\hat{\sigma}_{(i)}^2 = \|(I-H')\varepsilon\|^2$$